Derivative-Free Proofs of Gauss–Lucas and Marden via the Pandrosion Quotient

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Abstract

We give proofs of the Gauss–Lucas theorem and Marden's theorem that avoid the derivative $P'(z)$ entirely, using only the Pandrosion difference quotients $Q(\alpha_k, z) = \prod_{j \neq k}(z - \alpha_j)$. The key identity $P'(z) = \sum_{k=1}^{d} Q(\alpha_k, z)$ decomposes the derivative into *root contributions*, each of which is a polynomial that "encodes" the influence of root α_k on the critical structure of P. We introduce *Pandrosion influence weights* $\lambda_k(\beta) = |Q(\alpha_k, \beta)/(\beta - \alpha_k)|^2 / \sum_j |\cdots|^2$ that express each critical point as a weighted barycentre of the roots. As a byproduct, we prove a *centroid identity* for the roots of $Q(a, \cdot)$ and reinterpret Marden's theorem (foci of the Steiner inellipse) in pure Pandrosion language.

1 The Pandrosion decomposition of P'

Theorem 1.1 (Derivative as Pandrosion sum)**.** *Let $P(z) = \prod_{k=1}^{d}(z - \alpha_k)$ be a monic polynomial of degree d. Then:*

$$P'(z) \;=\; \sum_{k=1}^{d} Q(\alpha_k, z), \qquad Q(\alpha_k, z) := \prod_{\substack{j=1 \\ j \neq k}}^{d} (z - \alpha_j). \tag{1}$$

Proof. The product rule gives $P'(z) = \sum_k \prod_{j \neq k}(z - \alpha_j)$, which is (1). Alternatively, the difference quotient $Q(\alpha_k, z) = (P(z) - P(\alpha_k))/(z - \alpha_k) = P(z)/(z - \alpha_k)$ (since $P(\alpha_k) = 0$) gives $\sum_k P(z)/(z - \alpha_k) = P(z)\sum_k 1/(z - \alpha_k) = P(z) \cdot P'(z)/P(z) = P'(z)$. □

Remark 1.2. The identity (1) does *not* invoke the notion of a derivative. The right-hand side involves only the polynomial products $\prod_{j \neq k}(z - \alpha_j)$, which are the Pandrosion quotients $Q(\alpha_k, z) = (P(z) - P(\alpha_k))/(z - \alpha_k)$ evaluated at the roots. The derivative P' is merely a *name* for the sum $\sum_k Q(\alpha_k, z)$.

2 Derivative-free proof of Gauss–Lucas

Definition 2.1 (Pandrosion influence)**.** For a polynomial $P = \prod_k (z - \alpha_k)$ with simple roots and a point $z \notin \{\alpha_1, \ldots, \alpha_d\}$, the **Pandrosion influence** of root α_k on z is:

$$I_k(z) := \frac{Q(\alpha_k, z)}{z - \alpha_k} = \prod_{j \neq k} \frac{z - \alpha_j}{z - \alpha_k}. \tag{2}$$

Note: $\sum_k I_k(z) = P'(z)/P(z)$, the logarithmic derivative.

Theorem 2.2 (Gauss–Lucas, Pandrosion form). *Let $P(z) = \prod_{k=1}^{d}(z - \alpha_k)$ with $d \geq 2$ and let β be a zero of $\sum_k Q(\alpha_k, \cdot)$ with $P(\beta) \neq 0$. Then β is a convex combination of $\alpha_1, \ldots, \alpha_d$:*

$$\beta = \sum_{k=1}^{d} \lambda_k \, \alpha_k, \qquad \lambda_k = \frac{|\beta - \alpha_k|^{-2}}{\displaystyle\sum_{j=1}^{d} |\beta - \alpha_j|^{-2}} \in (0, 1]. \tag{3}$$

In particular, β lies in the convex hull of $\{\alpha_1, \ldots, \alpha_d\}$.

Proof. The hypothesis gives $\sum_k Q(\alpha_k, \beta) = 0$, i.e. $\sum_k \prod_{j \neq k}(\beta - \alpha_j) = 0$. Since $P(\beta) \neq 0$, dividing by $P(\beta)$:

$$\sum_{k=1}^{d} \frac{1}{\beta - \alpha_k} = 0.$$

Taking complex conjugates:

$$\sum_{k=1}^{d} \frac{1}{\overline{\beta - \alpha_k}} = 0,$$

and multiplying the k-th term by $(\beta - \alpha_k)/(\beta - \alpha_k)$:

$$\sum_{k=1}^{d} \frac{\beta - \alpha_k}{|\beta - \alpha_k|^2} = 0.$$

Separating β:

$$\beta \sum_{k=1}^{d} |\beta - \alpha_k|^{-2} = \sum_{k=1}^{d} \alpha_k \, |\beta - \alpha_k|^{-2}.$$

Setting $\lambda_k = |\beta - \alpha_k|^{-2} / \sum_j |\beta - \alpha_j|^{-2}$ gives $\beta = \sum_k \lambda_k \alpha_k$ with $\lambda_k > 0$ and $\sum_k \lambda_k = 1$. $\square$

Remark 2.3 (Derivative-free content). The proof uses:

1. The Pandrosion quotients $Q(\alpha_k, z) = \prod_{j \neq k}(z - \alpha_j)$ no derivative;
2. The condition $\sum_k Q(\alpha_k, \beta) = 0$ identifies critical points without saying "$P'(\beta) = 0$";
3. Complex conjugation and positivity of $|\beta - \alpha_k|^{-2}$.

The word "derivative" appears nowhere. The Gauss–Lucas theorem is a statement about where the Pandrosion sum $\sum_k Q(\alpha_k, z)$ can vanish.

Definition 2.4 (Pandrosion influence weights). At a zero β of $\sum_k Q(\alpha_k, \cdot)$, the weight

$$\lambda_k(\beta) = \frac{|\beta - \alpha_k|^{-2}}{\sum_j |\beta - \alpha_j|^{-2}} \tag{4}$$

is the **Pandrosion influence weight** of root α_k on β. It satisfies $\lambda_k \in (0, 1]$, $\sum \lambda_k = 1$, and measures how strongly root α_k "attracts" the critical point β. Nearby roots exert stronger influence: $\lambda_k \to 1$ as $\beta \to \alpha_k$ (forcing β toward α_k).

3 The Pandrosion centroid theorem

Theorem 3.1 (Centroid of $Q(a, \cdot)$ zeros)**.** *Let $P(z) = \prod_k (z - \alpha_k)$ be monic of degree d, and let $a \notin \{\alpha_1, \ldots, \alpha_d\}$. The polynomial $Q(a, z) = (P(z) - P(a))/(z - a)$ has degree $d - 1$ with roots $\beta_1, \ldots, \beta_{d-1}$ (the $d - 1$ other preimages of $P(a)$: $P(\beta_j) = P(a)$, $\beta_j \neq a$). Their centroid is:*

$$\frac{1}{d-1} \sum_{j=1}^{d-1} \beta_j = \frac{\sum_k \alpha_k - a}{d-1}. \tag{5}$$

Proof. $Q(a, z)$ is a monic polynomial of degree $d-1$ (the leading coefficient of P is 1, and dividing $(P(z) - P(a))$ by $(z - a)$ preserves monicness). Its sub-leading coefficient is the coefficient of z^{d-2}.

Write $P(z) = z^d + c_{d-1} z^{d-1} + \cdots$ with $c_{d-1} = -\sum_k \alpha_k$. By polynomial long division:

$$Q(a, z) = z^{d-1} + (a + c_{d-1})\, z^{d-2} + \cdots$$

By Vieta's formula, $\sum_j \beta_j = -(a + c_{d-1}) = \sum_k \alpha_k - a$. □

Remark 3.2. The centroid of the Q-zeros is a linear function of the anchor a: as a varies, the centroid traces the line from $(\sum \alpha_k)/(d - 1)$ (when $a = 0$) toward $-\infty$ (when $a \to +\infty$). In contrast, the *centroid of the critical points* (zeros of P') is always $(\sum \alpha_k)/d$ (by Vieta for P', which is independent of any anchor).

4 Marden's theorem via Pandrosion

Theorem 4.1 (Marden, 1945; Pandrosion formulation)**.** *Let $P(z) = (z - \alpha)(z - \beta)(z - \gamma)$ be a cubic with distinct roots forming a triangle $\triangle = \alpha\beta\gamma$. The two zeros of the Pandrosion sum*

$$Q(\alpha, z) + Q(\beta, z) + Q(\gamma, z) = (z - \beta)(z - \gamma) + (z - \alpha)(z - \gamma) + (z - \alpha)(z - \beta(= 0$$

are the foci of the unique ellipse inscribed in $\triangle$ and tangent to each side at its midpoint (the Steiner inellipse*).*

Proof. Expanding the Pandrosion sum:

$$\begin{aligned} Q(\alpha, z) + Q(\beta, z) + Q(\gamma, z) &= 3z^2 - 2(\alpha + \beta + \gamma)z + (\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= 3\left(z - \frac{\alpha + \beta + \gamma}{3}\right)^2 - \frac{(\alpha + \beta + \gamma)^2 - 3(\alpha\beta + \alpha\gamma + \beta\gamma)}{3}. \end{aligned}$$

Setting this to zero:

$$z = \frac{\alpha + \beta + \gamma}{3} \pm \frac{1}{3}\sqrt{(\alpha + \beta + \gamma)^2 - 3(\alpha\beta + \alpha\gamma + \beta\gamma)}.$$

The centroid is $(\alpha + \beta + \gamma)/3$, and the two zeros are symmetric about it. Siebeck (1864) and Marden (1945) proved that these are precisely the foci of the Steiner inellipse of $\triangle$.

The Pandrosion interpretation: each vertex contributes a quadratic "influence field" $Q(\alpha_k, z)$, and the critical points are where these three fields cancel. By symmetry, the cancellation points are the foci of the maximal inscribed ellipse. □

5 Higher-order Gauss–Lucas via Pandrosion

Theorem 5.1 (Higher-order derivative decomposition). *For $k \leq d$:*

$$\frac{P^{(k)}(z)}{k!} = e_k\big(Q(\alpha_1, z), \ldots, Q(\alpha_d, z)\big), \tag{6}$$

where e_k is the k-th elementary symmetric polynomial.

Proof. From $P(z) = \prod_j (z - \alpha_j)$, take $\log P = \sum \log(z - \alpha_j)$ and differentiate k times. The k-th logarithmic derivative involves sums over k-subsets; multiplying by P and collecting terms gives the elementary symmetric function of the cofactors $Q(\alpha_j, z) = P(z)/(z - \alpha_j)$. □

Corollary 5.2 (Iterated Gauss–Lucas). *The zeros of $e_k(Q(\alpha_1, z), \ldots, Q(\alpha_d, z))$ lie in the convex hull of $\{\alpha_1, \ldots, \alpha_d\}$ for every $k = 1, \ldots, d-1$. This is the classical iterated Gauss–Lucas theorem, expressed without derivatives.*

6 Conclusion

The Pandrosion quotient $Q(\alpha_k, z) = \prod_{j \neq k}(z - \alpha_j)$ provides a *derivative-free* language for the critical structure of polynomials:

Classical	**Pandrosion**
$P'(z)$	$\sum_k Q(\alpha_k, z)$
$P'(\beta) = 0$	$\sum_k Q(\alpha_k, \beta) = 0$
$P^{(k)}(z)/k!$	$e_k(Q(\alpha_1, z), \ldots, Q(\alpha_d, z))$
Gauss–Lucas	$\sum Q = 0 \Rightarrow \beta \in \mathrm{conv}\{\alpha_k\}$
Influence of α_k	$\lambda_k = \lvert\beta - \alpha_k\rvert^{-2} / \sum \lvert \cdots \rvert^{-2}$
Marden (cubics)	$Q(\alpha, z) + Q(\beta, z) + Q(\gamma, z) = 0$

All proofs are elementary and use only root products, never the formal differentiation operator. The *Pandrosion influence weights* provide a quantitative refinement of Gauss–Lucas: each critical point is attracted toward nearby roots with explicit strength $\lambda_k \propto |\beta - \alpha_k|^{-2}$.

References

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